

AI FAQ ASSISTANT API

2. Requirement Analysis Phase

2.1 Functional Requirements

The system should provide the following functions:

1. Question Input

The user/application should be able to send a question.

2. Question Processing

The API should process the received question.

3. FAQ Search

The system should search the available FAQ/knowledge base.

4. Answer Generation

The system should identify or generate an appropriate answer.

5. API Response

The answer should be returned in a structured format such as JSON.

6. Error Handling

The system should provide an appropriate message when:

- Question is empty.
- Question cannot be understood.
- No relevant FAQ is found.
- Server/API error occurs.

2.2 Non-Functional Requirements

Performance

The API should provide responses within a reasonable time.

Security

The API should protect sensitive information and validate incoming requests.

Reliability

The system should handle invalid requests without crashing.

Scalability

The application should support increasing numbers of users and questions.

Maintainability

The code should be organized so that future developers can easily modify it.

Usability

The API should be simple to use and integrate.

2.3 Software Requirements

- Operating System: Windows 10/11, macOS, or Linux (supports cross-platform operations).
- Node.js (v18 or above): Runtime ecosystem running server logic and routing infrastructure.
- npm (v9 or above): Node package manager to control operational dependencies.
- Express.js (v4/v5 Ecosystem): Lightweight routing web framework to construct backend RESTful entry points.
- MongoDB / Mongoose ODM: Document NoSQL engine storing application users and curated data entities.
- Postman / ThunderClient: API verification toolkit to assert schema validation responses across administrative route shields.
- Code Editor: Visual Studio Code or similar IDE environment.

2.3 Hardware Requirements

- Processor: Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent.
- RAM: 8 GB minimum (16 GB recommended for concurrent instances of MongoDB processing layers).
- Storage: 1 GB of available disk workspace.